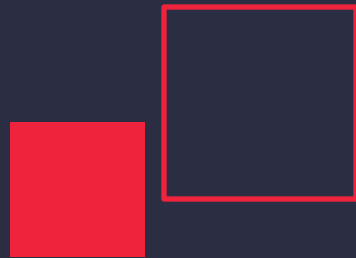

Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs

Why Software Tests Fail and Hardware Attestation Wins

arxiv.org/abs/2504.04715



The Model Substitution Problem: Economic Incentives Drive Covert Swaps

1

Smaller Models

e.g., 8B instead of 70B parameters

2

Quantized Variants

FP8/INT8 instead of FP16 precision

3

Fine-tuned Versions

Updated or modified weights

4

Different Families

Entirely different architectures

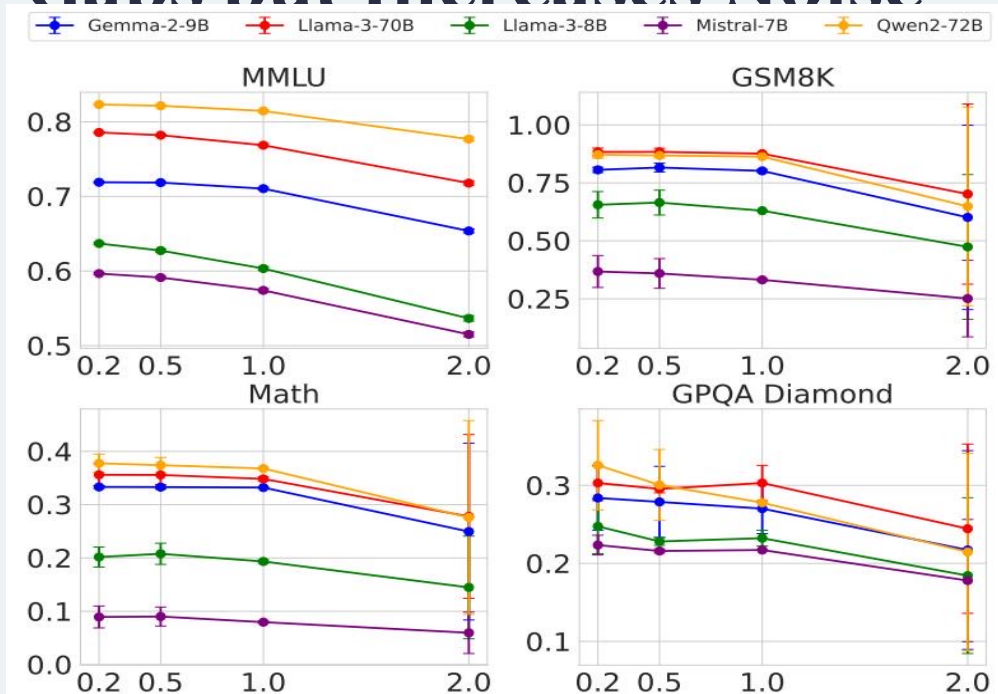
Hypothesis test: distinguish $P_{\text{spec}}(y|x)$ from $P_{\text{actual}}(y|x)$

FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69	61.14	20.65	22.62
<i>Llama-3-8B-Instruct-FP8</i>	<i>62.43</i>	<i>60.90</i>	<i>14.91</i>	<i>20.14</i>
Llama-3-70B-Instruct	78.05	88.06	35.69	29.60
<i>Llama-3-70B-Instruct-FP8</i>	<i>77.88</i>	<i>87.35</i>	<i>35.75</i>	<i>33.12</i>
Gemma-2-9b-it	71.86	81.80	33.41	28.64
<i>Gemma-2-9b-it-FP8</i>	<i>71.92</i>	<i>79.41</i>	<i>32.53</i>	<i>27.81</i>
Qwen2-72B-Instruct	82.18	86.72	37.39	29.93
<i>Qwen2-72B-Instruct-FP8</i>	<i>81.98</i>	<i>86.82</i>	<i>37.67</i>	<i>31.08</i>
Mistral-7B-Instruct-v0.3	59.15	35.90	8.94	21.60
<i>Mistral-7B-Instruct-v0.3-FP8</i>	<i>58.77</i>	<i>32.20</i>	<i>7.68</i>	<i>22.72</i>

Performance drops often within error bars — detection extremely challenging

Higher Temperature Amplifies Performance Gaps but Increases Noise



Source: arxiv.org/abs/2504.04715, Figure 1

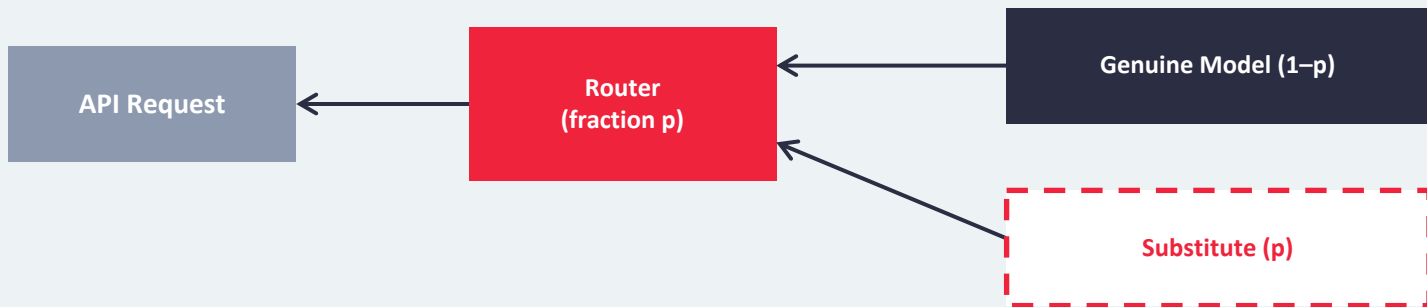
- Temperature scaling experiments tested $T = 0.2, 0.5, 1.0, 2.0$
- Higher temperatures degrade all models' performance
- The gap between original and FP8 substitute narrows or becomes noisy
- Statistical detection is further obscured by temperature variation

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

- Same Llama-3-8B model produces different log probabilities
- Variation across software: transformers 4.40 vs 4.50, vLLM 0.6.2 vs 0.8.2
- Variation across hardware: A100 vs H100 GPUs
- Auditor cannot distinguish infrastructure differences from actual substitution

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive



Randomized Substitution

Provider routes a fraction p of queries to a cheaper model. As $p \rightarrow 0$, the mixed output distribution converges to the specified model's distribution, requiring exponentially more queries ($O(1/p^2)$) to detect the substitution with statistical confidence.

Benchmark Evasion

Provider detects known benchmark prompts (e.g., MMLU, HumanEval) and routes them to the genuine model, while serving cheaper substitutes for regular user traffic. This defeats benchmark-based auditing without affecting measured performance.

Software-Only Auditing Methods Are Fundamentally Unreliable

Method Type	Weakness	Practical Limitation
Text-Based Statistical Tests (e.g., MMD)	Require impractically large query budgets	Fail against subtle or randomized substitutions; sample complexity grows exponentially as $p \rightarrow 0$
Log-Probability Fingerprinting	Defeated by inference nondeterminism	Software versions and GPU hardware cause variation indistinguishable from substitution
Adversarial Countermeasures	Providers can actively evade audits	Benchmark evasion routes known prompts to genuine model; randomized routing at low p is undetectable

All software-only approaches are fundamentally query-prohibitive

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

Trusted Execution Environments (TEEs) are hardware-secured enclaves that provide cryptographic attestation of the exact code and data being executed. With TEEs—such as NVIDIA confidential computing on H100 GPUs—providers can prove that specific model weights are loaded and correct inference code is running, with only modest performance overhead. Unlike zero-knowledge proofs, which remain computationally prohibitive for large models, TEEs are practical and deployable today.

Software Tests

- ✗ Unreliable
- ✗ Query-prohibitive
- ✗ Evadable

ZKPs

- ✓ Theoretically sound
- ✗ Computationally prohibitive
- ✗ Not practical for LLMs

TEEs

- ✓ Provable
- ✓ Efficient
- ✓ Robust
- ✓ Deployable today

Key Takeaways: Close the Verification Gap with Hardware Attestation

- 01 Model substitution is economically motivated and practically undetectable via software
- 02 FP8 quantization reduces costs significantly while barely affecting benchmark scores
- 03 Inference nondeterminism across software versions and GPU hardware defeats log-probability fingerprinting
- 04 Randomized substitution and benchmark evasion further undermine all software-only detection
- 05 Trusted Execution Environments are the only currently viable path to verifiable LLM inference

 Demand cryptographic attestation from your LLM API providers.